The 50 Most Difficult Injuries to Detect

A Clinical Research Brief for Musculoskeletal Triage & Differential Reasoning

Prepared for FisioterapIA clinical reasoning validation · 31 July 2026

Abstract

A large share of diagnostic error in musculoskeletal care is not “rare disease ignorance” but pattern capture failure: the patient’s complaint is anatomically downstream of the true lesion, initial imaging is falsely normal, neurologic signs are subtle, or a dangerous medical mimic occupies an orthopedic niche. This brief synthesates fifty high-miss-risk presentations spanning referred pain, occult fractures, peripheral and root neuropathies, tendon ruptures that still allow walking, inflammatory and infectious red flags, and visceral/ cardiac emergencies. For each entity we state the true pathology, the cognitive trap, the usual mislabel, and an ordered detection sequence clinicians (and clinical AI systems) should run before locking onto the obvious local diagnosis.

Method & scope

Selection prioritized (1) frequency of mislabeling in primary MSK pathways, (2) severity of harm if missed, and (3) availability of concrete bedside discriminators. Sources informing this synthesis include standard orthopedic/sports medicine differential frameworks, neurologic localization principles, Ottawa and stress-fracture imaging pathways, and published “missed diagnosis” patterns in emergency and outpatient MSK care. This document is educational and for clinical-reasoning validation; it is not a practice guideline and does not replace individualized clinical judgment, local protocols, or emergency pathways.

Cross-cutting detection principles

- Ask whether the local diagnosis explains ALL key findings (neurologic distribution, night pain, hop failure, systemic signs).
- Screen the joint above and the spine for referred-pain maps (hip!’knee, neck!’shoulder/elbow, lumbar!’foot).
- Treat normal early radiographs as provisional when bone pain physiology is classic.
- Autonomic, vascular, saddle, febrile, or cord signs outrank orthopedic tests—escalate first.
- Failed standard care for a “common” diagnosis is a signal to reopen the differential, not to intensify the same plan indefinitely.

Contents

Injuries are ordered by clinical detection difficulty and real-world miss risk—not by anatomical region. Each entry includes the true pathology, why it deceives, what it mimics, and a stepwise detection pathway.

1. C7 Radiculopathy Mimicking Lateral Epicondylitis
2. Hip Osteoarthritis Referred to the Knee
3. Femoral Neck Stress Fracture
4. Avascular Necrosis of the Femoral Head
5. Radial Tunnel Syndrome
6. Posterior Interosseous Nerve (PIN) Syndrome
7. Thoracic Outlet Syndrome (TOS)
8. SLAP Lesion (Superior Labrum)
9. Occult Scaphoid Fracture
10. Tarsal Tunnel Syndrome
11. S1 Radiculopathy Mimicking Plantar Fasciitis
12. Partial Achilles Tendon Rupture
13. L4 Radiculopathy Mimicking Meniscal Injury
14. Pectoralis Major Rupture
15. Myocardial Ischemia Presenting as Left Shoulder Pain
16. Sacroiliac Joint Pain vs Lumbar Disc Pain
17. Deep Gluteal / Piriformis Syndrome
18. Diaphragmatic / Visceral Referred Scapular Pain
19. Ulnar Neuropathy at the Elbow vs Medial Epicondylitis
20. Inflammatory Back Pain (Axial Spondyloarthritis Pattern)
21. Early Cervical Myelopathy
22. Vascular Claudication vs Neurogenic Claudication
23. Deep Vein Thrombosis (DVT)
24. Chronic Exertional Compartment Syndrome (CECS)
25. Stress Fracture with Initially Normal Radiographs
26. Lisfranc Injury
27. Navicular Stress Fracture
28. Syndesmotic (High Ankle) Sprain
29. Cauda Equina Syndrome
30. Acute Compartment Syndrome
31. Occult Hip Fracture in the Elderly
32. Proximal Humerus Fracture vs Rotator Cuff Tear
33. Distal Biceps Tendon Rupture
34. Extensor Mechanism Rupture (Quad or Patellar Tendon)
35. Gluteus Medius / Minimus Tear vs Trochanteric Bursitis
36. Fifth Metatarsal Base / Jones Fracture
37. Posterior Tibial Tendon Dysfunction (PTTD)
38. Pronator Syndrome (Median Nerve Proximal)
39. Cubital Tunnel Syndrome
40. Femoroacetabular Impingement / Labral Tear
41. Cervical Radiculopathy Presenting as Shoulder Pain
42. Spinal Infection (Discitis / Osteomyelitis)
43. Pars Interarticularis Stress Injury (Spondylolysis)
44. Meniscal Root Tear
45. Posterolateral Corner (PLC) Injury
46. Common Peroneal Nerve Palsy vs L5 Radiculopathy
47. Proximal Hamstring Tendon Avulsion
48. Osteochondral Lesion of the Talus (OLT)
49. Upper Cervical Instability (Atlantoaxial / Craniovertebral)
50. Parsonage–Turner Syndrome (Neuralgic Amyotrophy)

The fifty injuries

1. C7 Radiculopathy Mimicking Lateral Epicondylitis

Referred / neurologic

What it actually is

Compression or irritation of the C7 nerve root (often C6–C7 disc), producing lateral elbow pain, middle-finger sensory change, and triceps/wrist-extensor weakness.

Why it is tricky

Pain localizes to the lateral elbow and worsens with gripping—exactly like tennis elbow. Patients and clinicians focus on the tendon and miss the neck.

Commonly mistaken for

Lateral epicondylitis / common extensor tendinopathy

Correct detection steps

- Ask about neck pain, stiffness, or symptoms that change with cervical rotation/extension.
- Map sensory symptoms: middle finger (C7) vs purely local epicondyle pain.
- Test triceps strength and reflex; compare wrist/finger extension endurance.
- Palpate the epicondyle: pure epicondylitis is exquisitely tender on the bone; radiculopathy often is not.
- Perform cervical screening (active ROM, Spurling-type compression if safe) before treating as isolated tennis elbow.
- If neurologic signs dominate, prioritize cervical work-up and specialist referral over local elbow injections alone.

Red flags

- Progressive weakness
- Bilateral symptoms
- Gait change (myelopathy overlap)

Imaging / confirmation

Cervical MRI if persistent neurologic deficit or failed conservative care; elbow imaging only if local tendon signs dominate.

2. Hip Osteoarthritis Referred to the Knee

Referred pain

What it actually is

Coxarthrosis (hip OA) referring pain along the obturator/femoral sensory pathways so the patient reports “knee pain” while the joint of origin is the hip.

Why it is tricky

Many patients never mention the hip. Knee X-rays may be normal, so care stays stuck on the wrong joint.

Commonly mistaken for

Meniscal injury, patellofemoral pain, local knee OA

Correct detection steps

- Always screen the hip when knee pain is atypical or imaging is normal.
- Ask about limp, difficulty putting on socks/shoes, and groin or buttock pain.
- Examine hip internal rotation in flexion—marked limitation/pain is a classic clue.
- Compare knee exam: if the knee is quiet but gait and hip ROM are abnormal, re-rank the hip first.
- Obtain hip radiographs (AP pelvis + lateral) when suspicion is moderate–high.

Imaging / confirmation

Hip X-ray; MRI if AVN or soft-tissue differential remains.

3. Femoral Neck Stress Fracture

Occult bone injury

What it actually is

Fatigue or insufficiency fracture of the femoral neck, common in runners and endurance athletes, or in low bone density.

Why it is tricky

Early pain feels like a “groin strain” or iliopsoas tendinitis. Initial X-rays are often normal.

Commonly mistaken for

Iliopsoas tendinopathy, adductor strain, hip flexor overuse

Correct detection steps

- Red-flag history: progressive groin pain in a runner, night pain, pain with impact that is escalating.
- Single-leg hop/hop test often impossible or severely painful—do not force if high suspicion.
- Log-roll and gentle IR/ER often provoke deep pain.
- Never treat as “psoas tendinitis” if night pain + inability to hop + progressive running pain.
- Urgent imaging pathway: MRI (preferred) or bone scan if MRI unavailable; protect weight-bearing pending results.

Red flags

- Night pain
- Inability to hop
- Rapid progression
- Female athlete triad / RED-S risk

Imaging / confirmation

MRI gold standard early; X-ray may be falsely reassuring.

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4. Avascular Necrosis of the Femoral Head

Occult bone / vascular

What it actually is

Osteonecrosis of the femoral head from interrupted blood supply (steroids, alcohol, trauma, idiopathic), progressing to collapse if missed.

Why it is tricky

Presents as deep hip pain without trauma and preserved strength, so it is labeled “tendinopathy” or “bursitis.”

Commonly mistaken for

Hip flexor/adductor tendinopathy, GTPS, early OA

Correct detection steps

- Ask about corticosteroid use, alcohol excess, sickle cell disease, prior hip trauma.
- Deep groin pain + night pain + painful restricted internal rotation with relatively preserved strength.
- Do not stop at soft-tissue diagnosis if risk factors or night pain exist.
- Order MRI early (more sensitive than X-ray in early stages).
- Urgent orthopedic referral once confirmed or strongly suspected.

Imaging / confirmation

MRI; X-ray later stages (crescent sign, collapse).

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5. Radial Tunnel Syndrome

Peripheral nerve

What it actually is

Compression of the posterior interosseous / radial nerve in the radial tunnel (arcade of Frohse region), causing lateral forearm/elbow pain.

Why it is tricky

Overlaps almost perfectly with tennis elbow territory, but tenderness is distal to the epicondyle and local epicondyle care fails.

Commonly mistaken for

Lateral epicondylitis

Correct detection steps

- Palpate: maximal tenderness ~3–5 cm distal to the lateral epicondyle, not on the epicondyle itself.
- Pain with resisted middle-finger extension and/or resisted supination with elbow extended.
- Look for weak finger/wrist extension endurance without clear epicondyle tenderness.
- If “tennis elbow” fails standard care, reclassify as radial tunnel / PIN spectrum.
- Neurophysiologic studies may be normal—diagnosis is often clinical.

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6. Posterior Interosseous Nerve (PIN) Syndrome

Peripheral nerve

What it actually is

Motor neuropathy of the PIN causing inability to extend the fingers/thumb, often with little pain and normal sensation.

Why it is tricky

Absence of pain and preserved sensation make clinicians look for tendon rupture or “non-organic” weakness.

Commonly mistaken for

Extensor tendon rupture, C7 radiculopathy, stroke (rarely)

Correct detection steps

- Key triad: little/no pain + finger extension paralysis + normal sensation.
- Wrist may show radial deviation on extension (ECU spared pattern).
- Distinguish from tendon rupture by tenodesis effect and palpation of tendon continuity.
- Screen cervical roots and CNS signs to exclude central causes.
- Urgent hand/ortho referral; imaging/EMG as indicated.

Red flags

- Acute complete finger-drop
- Progressive motor loss

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7. Thoracic Outlet Syndrome (TOS)

Neurovascular

What it actually is

Compression of the brachial plexus and/or subclavian vessels at the thoracic outlet (scalenes, costoclavicular space, pectoralis minor).

Why it is tricky

Shoulder pain leads to a rotator-cuff pathway; vascular and positional clues are overlooked.

Commonly mistaken for

Rotator cuff tendinopathy, cervical radiculopathy, adhesive capsulitis

Correct detection steps

- Ask about whole-arm paresthesias, backpack/overhead aggravation, cold hand, color change, swelling.
- Reproduce symptoms with arms elevated / Roos-type endurance and postural assessment.

- Compare cuff testing: if cuff strength is near normal but neurovascular symptoms dominate, raise TOS.
- Rule out cervical radiculopathy and true cuff tear in parallel.
- Refer to specialist experienced in TOS; duplex/EMG selectively.

Red flags

- Acute arterial ischemia
- Significant venous swelling (effort thrombosis)

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8. SLAP Lesion (Superior Labrum)

Intra-articular shoulder

What it actually is

Tear of the superior labrum anterior–posterior, often in throwers/overhead athletes, involving the biceps anchor.

Why it is tricky

Deep pain and clicks are blamed on the rotator cuff; strength often remains nearly normal.

Commonly mistaken for

Rotator cuff tendinopathy, biceps tendinopathy, internal impingement

Correct detection steps

- Sport history: throwing, overhead, peel-back mechanism.
- Deep click/catch/lock with near-normal rotator cuff strength.
- Combine history with provocative labral tests; interpret cautiously (no single test is definitive).
- If conservative care for “cuff” fails in a thrower with mechanical symptoms, image for labrum.
- MR arthrogram often preferred over standard MRI for SLAP.

Imaging / confirmation

MR arthrogram; correlate surgically only when clinically indicated.

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9. Occult Scaphoid Fracture

Occult bone injury

What it actually is

Fracture of the scaphoid after FOOSH; initial radiographs frequently miss it because of delayed visibility.

Why it is tricky

Labeled “wrist sprain.” Missed fractures risk nonunion and AVN of the proximal pole.

Commonly mistaken for

Wrist sprain, De Quervain (without trauma history)

Correct detection steps

- Any FOOSH + snuffbox tenderness = scaphoid until proven otherwise.
- Add axial thumb load pain and scaphoid tubercle tenderness.
- If X-ray negative but clinical suspicion high: immobilize in thumb spica and re-image or MRI/CT.
- Never discharge “sprain” with classic signs and normal first X-ray without a safety-net plan.

Red flags

- Persistent snuffbox pain at 10–14 days

Imaging / confirmation

Repeat X-ray, MRI (early), or CT.

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10. Tarsal Tunnel Syndrome

Peripheral nerve

What it actually is
Tibial nerve compression in the tarsal tunnel behind the medial malleolus, causing plantar burning/paresthesia.

Why it is tricky
Plantar pain is almost automatically called plantar fasciitis.

Commonly mistaken for
Plantar fasciitis, Baxter's nerve entrapment, S1 radiculopathy

Correct detection steps
- Burning/tingling (not just first-step mechanical heel pain).
- Night pain and medial ankle pain; positive Tinel behind medial malleolus.
- Palpate plantar fascia origin—may be relatively quiet compared with classic fasciitis.
- Screen lumbar S1 so you do not miss radiculopathy.
- Consider EMG/US/MRI for space-occupying lesions if persistent.

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11. S1 Radiculopathy Mimicking Plantar Fasciitis

Referred / neurologic

What it actually is
S1 nerve-root irritation referring pain into the heel/plantar foot with sensory and reflex change.

Why it is tricky
Patients present only with "heel pain"; lumbar symptoms may be mild or forgotten.

Commonly mistaken for
Plantar fasciitis, fat-pad syndrome

Correct detection steps
- Ask specifically about low-back pain and leg symptoms.
- Check calf strength (heel raise endurance) and Achilles reflex.
- Map sensory change along S1.
- If neuro signs present, treat as radiculopathy work-up—not only fascia stretching.

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12. Partial Achilles Tendon Rupture

Tendon — incomplete tear

What it actually is
Incomplete tear of the Achilles; patient may still walk, so complete-rupture algorithms are not triggered.

Why it is tricky
Walking ability and a doubtful Thompson test lead to "tendinitis."

Commonly mistaken for
Achilles tendinopathy, calf strain

Correct detection steps
- History of pop/snap + localized defect tenderness.
- Single-leg heel raise often impossible even if walking is possible.
- Thompson may be equivocal—do not rely on it alone.
- Urgent ultrasound or MRI; early ortho/sports referral.
- Protect from full loading until tear extent is known.

Imaging / confirmation
Ultrasound or MRI.

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13. L4 Radiculopathy Mimicking Meniscal Injury

Referred / neurologic

What it actually is

L4 root irritation referring to the medial knee with quadriceps/reflex change.

Why it is tricky

Knee pain + medial paresthesia is attributed to meniscus; lumbar clues are soft.

Commonly mistaken for

Medial meniscus tear, MCL injury

Correct detection steps

- Ask about lumbar pain and medial-leg tingling.
- Test patellar reflex and knee-extension strength.
- If knee mechanical signs are weak but neuro signs exist, prioritize lumbar screening.
- Avoid unnecessary knee MRI-first pathways when the root explains the picture.

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14. Pectoralis Major Rupture

Muscle–tendon rupture

What it actually is

Tear of the pectoralis major (often at the tendon) during bench press or similar eccentric load.

Why it is tricky

Pain is felt in the shoulder/axilla and labeled “shoulder strain,” delaying surgical windows for complete tears.

Commonly mistaken for

Rotator cuff tear, biceps injury, AC sprain

Correct detection steps

- Bench-press / pec-deck mechanism + axillary ecchymosis is highly suggestive.
- Weakness of adduction/internal rotation with relatively preserved glenohumeral motion.
- Asymmetry of anterior axillary fold; loss of pec contour.
- Urgent MRI and orthopedic referral for complete tears (timing matters).

Imaging / confirmation

MRI chest wall / shoulder protocol for pec major.

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15. Myocardial Ischemia Presenting as Left Shoulder Pain

Medical emergency (referred)

What it actually is

Cardiac ischemia referring to the left shoulder/arm via shared visceral afferent pathways.

Why it is tricky

Patient seeks MSK care for “shoulder pain.” Movement-independent pain plus autonomic signs are missed.

Commonly mistaken for

Rotator cuff, AC joint, cervical referral

Correct detection steps

- If shoulder pain does NOT change with shoulder movement, stop MSK-only thinking.
- Ask about chest pressure, nausea, sweating, dyspnea, jaw/arm radiation.
- Autonomic signs (cold sweat, nausea) !’ emergency pathway immediately.
- Do not perform functional MSK tests as the priority action.
- Activate emergency medical services / ED.

Red flags

- Exertional chest pressure
- Diaphoresis
- Nausea

- Known CAD risk

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16. Sacroiliac Joint Pain vs Lumbar Disc Pain

Referred / regional differential

What it actually is

Nociception from the SI joint referring to the low back, buttock, and sometimes thigh—without a primary discogenic driver.

Why it is tricky

Everything “below L5” is labeled disc/sciatica; SI patterns are under-tested.

Commonly mistaken for

Non-specific LBP, L5–S1 disc, piriformis pain

Correct detection steps

- Pain with sit-to-stand, rolling in bed, prolonged sitting; buttock-predominant pain.
- Use a cluster of SI provocation tests (not one test alone).
- Screen neurologic status to avoid missing true radiculopathy.
- Consider inflammatory SI disease in younger patients with morning stiffness.

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17. Deep Gluteal / Piriformis Syndrome

Peripheral nerve entrapment

What it actually is

Sciatic nerve irritation in the deep gluteal space (piriformis and related structures).

Why it is tricky

Looks like lumbar radiculopathy or “simple gluteal strain.” Imaging of the spine may be nonspecific.

Commonly mistaken for

L5–S1 radiculopathy, hamstring tendinopathy, GTPS

Correct detection steps

- Buttock pain with sitting intolerance; sciatic-like distal symptoms with quieter lumbar findings.
- Palpation and stretch/contraction of deep external rotators may reproduce symptoms.
- Still rule out lumbar root compression—deep gluteal is a diagnosis of careful exclusion plus positive local signs.
- MRI lumbar ± pelvis when unclear; specialist referral if progressive deficit.

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18. Diaphragmatic / Visceral Referred Scapular Pain

Visceral referral

What it actually is

Irritation of the diaphragm or upper abdominal/thoracic viscera referring to the scapular region via phrenic/shared pathways (e.g., gallbladder, subdiaphragmatic processes).

Why it is tricky

Treated endlessly as scapular dyskinesis or cuff disease; MSK tests do not explain systemic/digestive links.

Commonly mistaken for

Rhomboid strain, scapular tendinopathy, cervical referral

Correct detection steps

- Scapular pain poorly related to shoulder motion.
- Ask about digestion, respiration, fever, abdominal symptoms.
- If MSK exam is unrevealing, escalate medical evaluation rather than more shoulder rehab alone.

Red flags

- Fever
- Jaundice
- Severe abdominal pain

- Dyspnea

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19. Ulnar Neuropathy at the Elbow vs Medial Epicondylitis

Peripheral nerve

What it actually is

Compression/irritation of the ulnar nerve in the cubital tunnel causing medial elbow pain plus ulnar-digit paresthesia.

Why it is tricky

Medial elbow pain is labeled golfer's elbow; sensory/motor ulnar signs are ignored.

Commonly mistaken for

Medial epicondylitis, UCL injury in throwers

Correct detection steps

- Ask about 4th/5th digit tingling and night symptoms.
- Tinel at cubital tunnel; elbow flexion test.
- Intrinsic weakness, clawing, or sensory loss !' nerve pathway, not only tendon.
- Palpate medial epicondyle: pure tendinopathy lacks ulnar distribution symptoms.

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20. Inflammatory Back Pain (Axial Spondyloarthritis Pattern)

Inflammatory vs mechanical

What it actually is

Inflammatory axial disease pattern (e.g., axial SpA) presenting as chronic low-back/buttock pain with inflammatory features.

Why it is tricky

Years of "mechanical LBP" labeling delay rheumatology referral and imaging.

Commonly mistaken for

Mechanical nonspecific LBP, SI mechanical pain

Correct detection steps

- Age of onset <40–45, morning stiffness >45 minutes, improvement with activity, night pain in second half of night.
- Ask about psoriasis, IBD, uveitis, family history.
- Do not endlessly prescribe only "core stability" without inflammatory screening.
- Refer rheumatology; HLA-B27 and MRI SI joints per guidelines.

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21. Early Cervical Myelopathy

Central neurologic

What it actually is

Spinal cord compression in the cervical canal causing hand clumsiness, gait imbalance, and often underestimated neck symptoms.

Why it is tricky

Early signs are subtle—"stiff neck" or hand numbness—until gait and dexterity clearly fail.

Commonly mistaken for

Cervical radiculopathy, carpal tunnel, peripheral neuropathy

Correct detection steps

- Ask about bilateral hand numbness, dropping objects, buttons/zippers difficulty, unsteady gait.
- Upper motor neuron signs: hyperreflexia, Hoffmann, clonus, Babinski, spastic gait.
- This is NOT routine mechanical neck pain—urgent specialist imaging pathway.

Red flags

- Gait ataxia
- Hand clumsiness

- Bowel/bladder change

Imaging / confirmation
Cervical MRI urgently when myelopathy suspected.

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22. Vascular Claudication vs Neurogenic Claudication

Vascular vs spine

What it actually is
Arterial insufficiency causing exertional leg pain that resolves promptly with standing rest (not necessarily flexion).

Why it is tricky
Walking-related leg pain is reflexively called “sciatica” or stenosis.

Commonly mistaken for
Neurogenic claudication from lumbar stenosis, radiculopathy

Correct detection steps
- Vascular: relief by standing still; neurogenic: often relief by sitting/flexing.
- Check pulses, skin, temperature, cardiovascular risk factors.
- Bike vs walk patterns can help differentiate in clinic.
- ABI / vascular referral when pulses diminished or risk high.

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23. Deep Vein Thrombosis (DVT)

Medical emergency

What it actually is
Lower-limb deep venous thrombosis; calf pain/swelling can mimic muscle tear.

Why it is tricky
Athletes and clinicians assume “tennis leg.” Missed DVT risks PE.

Commonly mistaken for
Medial gastrocnemius tear, soleus strain, cellulitis

Correct detection steps
- Wells score elements: swelling, warmth, risk factors (travel, surgery, OCP, cancer, prior DVT).
- No clear pop/mechanism + progressive unilateral swelling !’ think clot.
- Do not deep-massage or aggressive soft-tissue work on a suspected DVT.
- Urgent duplex ultrasound / ED pathway.

Red flags
- Sudden dyspnea/chest pain (PE)

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24. Chronic Exertional Compartment Syndrome (CECS)

Exertional syndrome

What it actually is
Reversible rise in intracompartmental pressure during exercise causing predictable distance-related pain and tightness.

Why it is tricky
Labeled shin splints; resting exam is often normal.

Commonly mistaken for
Medial tibial stress syndrome, tibial stress fracture

Correct detection steps
- Pain appears at a reproducible time/distance and eases after stopping.
- Sense of tightness/swelling in a compartment; possible transient numbness.
- Distinguish from stress fracture (focal bone tenderness, night pain).
- Confirm with post-exercise compartment pressure testing in specialist centers.

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25. Stress Fracture with Initially Normal Radiographs

Occult bone injury

What it actually is

Fatigue fracture of weight-bearing bone (tibia, metatarsals, pelvis, femoral neck, etc.) invisible on early X-ray.

Why it is tricky

Normal X-ray falsely reassures; athletes continue loading and complete the fracture.

Commonly mistaken for

Tendinopathy, MTSS, "growing pains" in adolescents

Correct detection steps

- Progressive focal bony pain, impact aggravation, night pain, hop pain.
- Point tenderness on bone, not diffuse soft tissue.
- If suspicion high: MRI despite normal X-ray; reduce load immediately.

Imaging / confirmation

MRI preferred; repeat X-ray later may show callus.

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26. Lisfranc Injury

Midfoot instability / fracture-dislocation

What it actually is

Injury to the tarsometatarsal complex; ranges from subtle ligamentous injury to fracture-dislocation.

Why it is tricky

Can look like a "foot sprain"; weight-bearing films needed; missed injuries cause chronic midfoot collapse.

Commonly mistaken for

Midfoot sprain, metatarsal contusion

Correct detection steps

- Mechanism: axial load on plantarflexed foot; plantar ecchymosis is a classic clue.
- Pain with midfoot squeeze / stress; inability to toe-raise comfortably.
- Weight-bearing bilateral X-rays; CT/MRI if still suspected.
- Urgent ortho referral—do not "walk it off."

Imaging / confirmation

Weight-bearing X-ray; CT/MRI for occult injury.

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27. Navicular Stress Fracture

Occult bone injury

What it actually is

Stress fracture of the tarsal navicular—high-risk location due to watershed blood supply.

Why it is tricky

Vague dorsal midfoot pain in runners; X-rays often normal early.

Commonly mistaken for

Extensor tendinopathy, midfoot sprain

Correct detection steps

- Insistent dorsal-medial midfoot pain in impact athletes.
- Pain over navicular ("N" spot); hop/push-off pain.
- High suspicion !' MRI/CT and non-weight-bearing protection pending diagnosis.
- Specialist management—high complication risk if missed.

Imaging / confirmation

MRI or CT; X-ray insensitive early.

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28. Syndesmotic (High Ankle) Sprain

Ligament — often under-graded

What it actually is

Injury to the tibiofibular syndesmosis, often with external rotation force.

Why it is tricky

Treated as ordinary lateral ankle sprain; recovery is much slower and instability persists.

Commonly mistaken for

Lateral ankle sprain (ATFL)

Correct detection steps

- Pain above the ankle joint line; external rotation mechanism.
- Squeeze test and external-rotation stress reproduce proximal pain.
- Assess for associated Maisonneuve (proximal fibula) injury.
- Imaging for diastasis; ortho if unstable.

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29. Cauda Equina Syndrome

Surgical emergency

What it actually is

Compression of the cauda equina causing saddle anesthesia, bowel/bladder dysfunction, and often bilateral leg symptoms.

Why it is tricky

Early stages may look like “bad sciatica”; embarrassment delays reporting of perineal/bladder symptoms.

Commonly mistaken for

Severe radiculopathy, mechanical LBP flare

Correct detection steps

- Mandatory questions: saddle numbness, urinary retention/incontinence, fecal incontinence, bilateral sciatica, sexual dysfunction.
- Any positive !’ emergency MRI and surgical referral NOW.
- Do not wait for “a few days of physio.”

Red flags

- Saddle anesthesia
- Urinary retention
- Bilateral motor loss

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30. Acute Compartment Syndrome

Surgical emergency

What it actually is

Acute rise in compartment pressure after fracture, crush, or reperfusion—threatens muscle and nerve viability.

Why it is tricky

Early pain is blamed on the fracture/contusion; classic “6 Ps” appear late.

Commonly mistaken for

Severe contusion, tight cast discomfort

Correct detection steps

- Pain out of proportion, pain on passive stretch, tense compartment.
- Paresthesia early; pulselessness is a LATE sign—do not wait for it.
- Emergency surgical fasciotomy pathway—minutes matter.

Red flags

- Pain out of proportion

- Tense swelling after trauma

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31. Occult Hip Fracture in the Elderly

Occult bone injury

What it actually is

Femoral neck or intertrochanteric fracture after low-energy fall; initial X-ray may be negative.

Why it is tricky

Patient "can move a little," so it is called bruise/strain; delayed diagnosis increases morbidity.

Commonly mistaken for

Soft-tissue contusion, GTPS flare

Correct detection steps

- Fall in older adult + inability to bear weight = fracture until cleared.
- Shortened/externally rotated limb when displaced; occult fractures may lack deformity.
- If X-ray negative but cannot walk: MRI (or CT) same-day pathway.

Imaging / confirmation

X-ray first; MRI if occult suspected.

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32. Proximal Humerus Fracture vs Rotator Cuff Tear

Occult bone / trauma

What it actually is

Fracture of the proximal humerus (common in osteoporotic falls) presenting with shoulder pain and weakness.

Why it is tricky

Weak elevation is attributed to cuff tear; fracture is missed without imaging.

Commonly mistaken for

Acute rotator cuff tear, dislocation (reduced)

Correct detection steps

- Age + fall + global pain/weakness !' radiograph before aggressive cuff testing.
- Ecchymosis down the arm is a useful clue.
- Never force PROM stress tests until fracture is excluded.

Imaging / confirmation

Shoulder X-ray series first-line.

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33. Distal Biceps Tendon Rupture

Tendon rupture

What it actually is

Avulsion of the distal biceps from the radial tuberosity, usually during heavy eccentric flexion/supination.

Why it is tricky

Elbow "strain" label; surgical delay worsens outcomes for complete tears.

Commonly mistaken for

Biceps strain, cubital bursitis, lateral elbow pain syndromes

Correct detection steps

- Pop during lift + antecubital pain + ecchymosis.
- Weakness especially of supination; Hook test / squeeze tests as trained.
- MRI or ultrasound confirmation; urgent ortho for complete ruptures.

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34. Extensor Mechanism Rupture (Quad or Patellar Tendon)

Tendon rupture

What it actually is

Rupture of the quadriceps tendon or patellar tendon disrupting active knee extension.

Why it is tricky

Patient may still walk with gait adaptation; swelling hides the gap.

Commonly mistaken for

Patellar dislocation, ACL injury, contusion

Correct detection steps

- Inability to perform straight-leg raise / active terminal extension.
- Palpable gap above or below patella; patella alta/baja.
- Urgent imaging and surgical referral—this is not “rest and ice” alone.

Imaging / confirmation

X-ray (patellar height); ultrasound/MRI for confirmation.

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35. Gluteus Medius / Minimus Tear vs Trochanteric Bursitis

Tendon — mislabeled bursitis

What it actually is

Insertional tear/tendinopathy of gluteus medius/minimus at the greater trochanter—often the true driver of “bursitis.”

Why it is tricky

Decades of “trochanteric bursitis” labeling; tears need different loading and sometimes repair.

Commonly mistaken for

Isolated bursitis, lumbar referral, hip OA

Correct detection steps

- Lateral hip pain, night pain lying on side, Trendelenburg/abduction weakness.
- Pain with resisted abduction and external rotation in sidelying.
- Ultrasound/MRI to grade tear if weakness is clear or injections fail.
- Rehab targets gluteal loading—not only “bursal anti-inflammatories.”

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36. Fifth Metatarsal Base / Jones Fracture

Occult / high-risk fracture

What it actually is

Fracture at zones of the proximal 5th metatarsal; Jones (zone 2) has higher nonunion risk.

Why it is tricky

Treated as ankle sprain; patient keeps playing.

Commonly mistaken for

Lateral ankle sprain, peroneal tendinopathy

Correct detection steps

- Point bone pain at base of 5th MT after inversion injury.
- Always palpate the 5th MT in “ankle sprains.”
- X-ray; classify zone; Jones often needs protected management / ortho.

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37. Posterior Tibial Tendon Dysfunction (PTTD)

Progressive tendon failure

What it actually is

Degeneration/failure of the tibialis posterior tendon leading to progressive adult-acquired flatfoot.

Why it is tricky

Early medial ankle pain is called “fasciitis” or sprain; deformity progresses quietly.

Commonly mistaken for

Plantar fasciitis, medial ankle sprain, tarsal tunnel

Correct detection steps

- Medial ankle/foot pain, “too many toes,” difficulty single-leg heel raise (heel fails to invert).
- Assess arch collapse and forefoot abduction.
- Early staging changes treatment (ortho/ortho-foot pathway).

38. Pronator Syndrome (Median Nerve Proximal)

Peripheral nerve

What it actually is

Median nerve compression in the proximal forearm (pronator teres region), overlapping carpal-tunnel-like sensory symptoms.

Why it is tricky

Automatically treated as carpal tunnel; night-only CTS pattern may be absent; forearm pain dominates.

Commonly mistaken for

Carpal tunnel syndrome, C6–C7 radiculopathy

Correct detection steps

- Forearm pain with repetitive pronation; sensory symptoms in median distribution.
- Provocative pronator maneuvers; negative or incomplete response to CTS-only care.
- EMG may help localize; still a clinical diagnosis often.

39. Cubital Tunnel Syndrome

Peripheral nerve

What it actually is

Ulnar nerve entrapment at the elbow with sensory ± motor loss in the ulnar hand.

Why it is tricky

Intermittent tingling blamed on “circulation” or neck; intrinsic weakness appears late.

Commonly mistaken for

Cervical C8–T1 radiculopathy, Guyon’s canal, medial epicondylitis

Correct detection steps

- 4–5 digit paresthesia, worse with prolonged elbow flexion (phone, driving).
- Elbow flexion test, Tinel at groove; assess intrinsics and clawing.
- Differentiate from cervical root with neck exam and reflex/myotome map.
- Activity modification early; EMG if surgery considered.

40. Femoroacetabular Impingement / Labral Tear

Intra-articular hip

What it actually is

Cam/pincer morphology with labral injury causing deep groin pain, clicking, and sitting intolerance.

Why it is tricky

Misdiagnosed as adductor strain or sports hernia for months.

Commonly mistaken for

Adductor tendinopathy, iliopsoas strain, inguinal disruption

Correct detection steps

- Deep groin pain, C-sign, pain with sitting/flexion, mechanical click.
- FADIR/FABER as screening—not definitive alone.
- X-ray for morphology; MR arthrogram for labrum when indicated.
- Exclude stress fracture/AVN when night pain or hop failure present.

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41. Cervical Radiculopathy Presenting as Shoulder Pain

Referred / neurologic

What it actually is

Cervical nerve-root pain referring to the shoulder girdle without prominent “neck complaint.”

Why it is tricky

Shoulder pathways (cuff, bursitis) absorb all attention.

Commonly mistaken for

Rotator cuff tendinopathy, subacromial pain syndrome

Correct detection steps

- Ask about neck movement aggravation and distal paresthesia.
- If cuff tests are inconsistent and cervical motions reproduce shoulder pain, raise cervical source.
- Spurling/cervical distraction as tolerated; neurologic screen of the limb.
- Image cervical spine when deficits persist—not only the shoulder.

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42. Spinal Infection (Discitis / Osteomyelitis)

Infection red flag

What it actually is

Infection of disc/vertebral body presenting as progressive back pain, often with insidious onset.

Why it is tricky

May lack fever early; attributed to mechanical strain, especially post-procedure or in IVDU/immunocompromise.

Commonly mistaken for

Mechanical LBP, vertebral compression fracture

Correct detection steps

- Risk: recent infection, surgery/injection, IV drug use, immunosuppression, diabetes.
- Constant pain, night pain, fever/chills, elevated inflammatory markers.
- Urgent labs (CRP/ESR/CBC) + MRI; do not delay for “failed physio.”

Red flags

- Fever
- IVDU
- Recent spinal procedure
- Neurologic deficit

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43. Pars Interarticularis Stress Injury (Spondylolysis)

Occult bone injury — adolescent athlete

What it actually is

Stress reaction/fracture of the pars, common in young athletes with repetitive extension (gymnastics, football, cricket bowling).

Why it is tricky

Called “lumbar strain”; plain films may miss early lesions.

Commonly mistaken for

Mechanical LBP, facet irritation, discogenic pain

Correct detection steps

- Adolescent + sport + extension-based pain, often unilateral.
- One-legged hyperextension pain is a clue (not pathognomonic).
- MRI (and sometimes CT for fracture detail) when suspicion high.
- Load management essential—early detection prevents nonunion/spondylolisthesis progression.

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44. Meniscal Root Tear

Intra-articular — subtle but severe

What it actually is

Avulsion/tear of the meniscal root that disables hoop stress function—biomechanically similar to total meniscectomy if missed.

Why it is tricky

May lack classic locking; MRI can be subtle; patients get “degenerative meniscus” labels.

Commonly mistaken for

Simple degenerative meniscal tear, early OA flare

Correct detection steps

- Middle-aged patient, deep joint-line pain, popping, subtle instability, rapid OA progression clues.
- Extrusion of meniscus on MRI is a warning sign.
- Specialist MRI review; early ortho discussion—repair windows matter.

Imaging / confirmation

High-quality MRI with attention to root attachments.

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45. Posterolateral Corner (PLC) Injury

Multi-ligament — often missed

What it actually is

Injury to LCL/popliteus/popliteofibular complex; often with ACL/PCL in high-energy trauma.

Why it is tricky

Focus stays on ACL; varus/rotatory instability is missed, and isolated ACL reconstruction fails.

Commonly mistaken for

Isolated ACL tear, lateral meniscus tear

Correct detection steps

- Hyperextension/varus mechanism; lateral/posterolateral pain.
- Dial test, varus stress at 0° and 30°, reverse pivot patterns as trained.
- Always examine PLC when ACL/PCL injured.
- MRI + specialist multi-ligament expertise.

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46. Common Peroneal Nerve Palsy vs L5 Radiculopathy

Peripheral vs root neurologic

What it actually is

Compression of the common peroneal nerve at the fibular neck causing foot drop and sensory change on the dorsum of the foot.

Why it is tricky

Foot drop is automatically “L5 radiculopathy”; local fibular compression (cast, habit sitting, trauma) is missed.

Commonly mistaken for

L5 radiculopathy, sciatic neuropathy, stroke (central)

Correct detection steps

- Foot drop + sensory map: peroneal sensory vs L5 dermatome differences; hip abduction often weak in L5, spared in pure peroneal.
- Palpate/Tinel at fibular neck; history of compression.
- EMG localizes lesion; image lumbar only when root more likely.
- Urgent protection from falls; address reversible compression.

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47. Proximal Hamstring Tendon Avulsion

Tendon avulsion

What it actually is

Avulsion of proximal hamstring tendons from the ischial tuberosity during forceful hip flexion with knee extension (water skiing, splits, sprint).

Why it is tricky

Called “high hamstring strain”; complete avulsions needing surgery are delayed.

Commonly mistaken for

Hamstring muscle belly strain, ischial bursitis, sciatica

Correct detection steps

- Sudden pop at buttock, ecchymosis, weakness of hip extension/knee flexion.
- Palpable gap near ischium; sitting pain severe.
- MRI to grade; early ortho for complete multi-tendon avulsions.

Imaging / confirmation

Pelvic MRI.

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48. Osteochondral Lesion of the Talus (OLT)

Intra-articular cartilage/bone

What it actually is

Cartilage ± underlying bone injury of the talar dome, often after ankle sprain.

Why it is tricky

Chronic “sprain that never healed”; plain films may miss lesions.

Commonly mistaken for

Chronic ankle instability, anterolateral impingement, peroneal tendinopathy

Correct detection steps

- Deep ankle pain, effusion, catching after prior sprain.
- If rehab for sprain fails, image for OLT.
- MRI preferred; CT for bone detail/surgical planning.

Imaging / confirmation

MRI; X-ray may show cystic change later.

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49. Upper Cervical Instability (Atlantoaxial / Craniovertebral)

High-risk cervical

What it actually is

Instability at C1–C2 or craniovertebral junction (rheumatoid arthritis, Down syndrome, trauma) threatening the cord/brainstem.

Why it is tricky

Presents as "neck pain/headache"; standard mechanical care is dangerous if instability exists.

Commonly mistaken for

Cervicogenic headache, mechanical neck pain, migraine

Correct detection steps

- Screen RA, connective tissue disorders, Down syndrome, significant trauma.
- Red flags: occipital headache, cord signs, cranial nerve symptoms, severe restricted rotation.
- Avoid end-range manipulation; urgent specialist imaging (flexion-extension / MRI/CT per protocol).

Red flags

- Neurologic long-tract signs
- RA + neck pain + myelopathy clues

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50. Parsonage–Turner Syndrome (Neuralgic Amyotrophy)

Inflammatory plexopathy

What it actually is

Acute brachial plexitis: severe shoulder/arm pain followed by patchy weakness and atrophy as pain eases.

Why it is tricky

Phase 1 looks like acute cuff tear or cervical radiculopathy; phase 2 weakness is blamed on "disuse."

Commonly mistaken for

Acute rotator cuff tear, C5–C6 radiculopathy, adhesive capsulitis

Correct detection steps

- Hyperacute severe pain (often night) followed days later by multifocal weakness (scapular winging, anterior interosseous, etc.).
- Patchy, non-myotomal pattern is a clue vs single-root disease.
- EMG after ~3 weeks; MRI plexus/cervical to exclude structural causes.
- Early recognition prevents futile "cuff tear surgery" pathways; pain control + specialist neurology/ortho.

How to use this brief for AI / clinical validation

- Present each case vignette without the title; require ranked differentials and an action (treat locally / image / refer urgently).
- Score whether the true injury appears in the top three hypotheses and whether red-flag pathways were triggered when indicated.
- Penalize "obvious local" diagnoses that ignore neurologic, nocturnal, vascular, or visceral features listed in detection steps.
- Pair this brief with a larger easy/intermediate/expert vignette battery to measure calibration, not only rare-case recall.

Disclaimer

Educational research brief for physiotherapy/MSK clinical reasoning. Not a substitute for licensed medical assessment, emergency care, or specialty guidelines. When in doubt about fracture, infection, cauda equina, compartment syndrome, DVT/PE, or cardiac ischemia, escalate immediately through appropriate urgent pathways.